

Solution Report

Dataset Structure

- R_1 : Contains 1000 tuples with $A_2 = 5$ and 1000 tuples with $A_2 = 7$.
- R_2 : Contains 1000 tuples with $A_2 = 5$ and 1000 tuples with $A_2 = 7$.
- $R_1 \bowtie R_2$:
 - The join on $A_2 = 5$ produces $1000 \times 1000 = 1,000,000$ tuples.
 - The join on $A_2 = 7$ produces $1000 \times 1000 = 1,000,000$ tuples.
 - **Total Intermediate Size**: $\sim 2,000,000$ tuples.
- R_3 : Mostly contains random values for A_3 that **do not match** the results from $R_1 \bowtie R_2$. The only matching path is via the specific “needle in the haystack” tuples (e.g., (2002, 8) in R_2 matching (8, 30) in R_3).

Comparison Results

We ran both algorithms on the generated dataset. The code can be found in [Solution.js](#). Note that we directly imported and reused the `generalizedLineJoin` implementation from Problem 2 and the `sequentialJoin` implementation from Problem 3 to ensure a fair comparison of the exact same logic.

Algorithm	Execution Time	Number of Results
Yannakakis (Problem 2)	~ 1.33 ms	1001
Sequential Join (Problem 3)	~ 372.71 ms	1001

(Note: Times are from a representative run. The difference is consistently orders of magnitude.)

Do they return the same results?

Yes. Output of P2 and P3 are identical. Both algorithms correctly identified the 1001 result tuples.

Performance Analysis

Yannakakis is significantly faster (over 200x faster in this test).

Why?

1. Sequential Join (Left-Deep):

- It blindly computes the full join $R_1 \bowtie R_2$ first.
- This generates an intermediate dataset of **~ 2 million tuples**.
- It then attempts to join these 2 million tuples with R_3 .

- Most of this work is wasted because the vast majority of these intermediate tuples have no match in R_3 . The algorithm pays the cost of generating and processing a huge intermediate relation.

2. Yannakakis Algorithm:

- **Reduction Phase:** It first filters the relations.
 - R_2 is reduced by R_3 : The 2000 tuples in R_2 with $A_3 \in [1, 2000]$ are checked against R_3 . Since R_3 mostly has $A_3 \geq 2002$, almost all tuples in R_2 are removed immediately!
 - **Key Pruning Fact:** The entire block of tuples with $A_2 = 7$ is completely eliminated because their A_3 values (1001-2000) never match R_3 . **Only tuples with attribute value A2 equal to 5 survive during semijoin pruning** (specifically the (5, 8) tuple), along with the single (2002, 8) tuple.
 - R_1 is then reduced by the tiny R'_2 .
- **Join Phase:** By the time the join happens, the relations are tiny. The “explosion” of 1000×1000 never happens because the participating tuples were filtered out *before* the join.
- This demonstrates the power of **semijoin reduction**: it avoids generating dangling intermediate results, ensuring $O(N + OUT)$ complexity rather than potentially $O(N^2)$ or worse for intermediate steps.